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B.Tech. Degree III Semester Supplementary Examination April 2019

IT/CS 15-1303 DISCRETE COMPUTATIONAL STRUCTURES
(2015 Scheme)

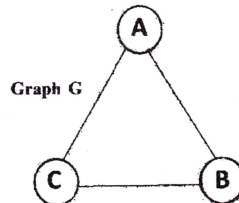
Time: 3 Hours

Maximum Marks: 60

PART A
(Answer *ALL* questions)

(10 × 2 = 20)

- I. (a) Prove that $A \times (B \cap C) = (A \times B) \cap (A \times C)$.
 (b) Prove that the relation $\leq$ is a partially ordered relation by defining it on the set of integers.
 (c) Prove that $p \vee \tilde{p}$ is a tautology and $p \wedge \tilde{p}$ is a contradiction.
 (d) Define asymptotic notations used for the time complexity of an algorithm.
 (e) How many 7 letter variable names can be formed from the alphabets 'a to i' with no letters repeated?
 (f) Draw a 3 regular graph with 4 vertices.
 (g) Give the memory representation for a directed graph using suitable example.
 (h) Draw the spanning trees of the given graph.



- (i) Define subgroup and monoid with example.
 (j) Let $A = \{1, 5, 6, 10, 15, 18, 24, 30, 60\}$ with partial ordering *divides*. Draw the Hasse diagram and find the maximal and minimal element of A.

PART B

(4 × 10 = 40)

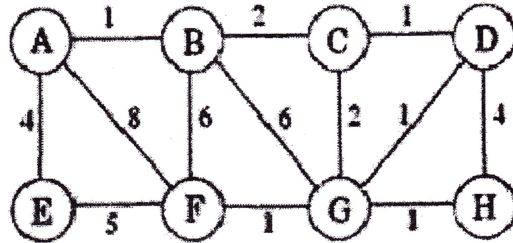
- II. (a) Write the contrapositive, converse and inverse of the expression $P \rightarrow Q, \tilde{P} \rightarrow Q, Q \rightarrow \tilde{P}$. (6)
 (b) Prove the validity of the given statements without using truth table "If I don't study hard, I will not pass this course. If I don't pass this course I cannot graduate this year. Therefore If I don't study hard, I won't graduate this year". (4)
- OR**
- III. (a) Prove by mathematical induction $1. 2 + 2. 3 + 3. 4 + \dots + n. (n + 1) = (1/3) n. (n + 1). (n + 2)$ (5)
 (b) Find the transitive closure of the relation $R = \{(a, b), (b, c), (c, d)\}$ on the set $A = \{a, b, c, d\}$ using Warshall's Algorithm. (5)

(P.T.O)

- IV. (a) Define algorithm and its best case, worst case and average case complexity. (5)
 (b) Show that in a group of 15 people, at least three were born on the same day of the week. (5)

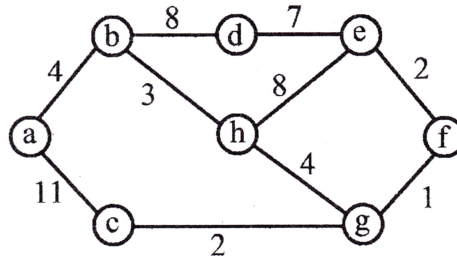
OR

- V. (a) Solve the recurrence equation $F_r - F_{r-1} - F_{r-2} = 0$ with initial conditions $F_0 = 1$ and $F_1 = 1$. (5)
 (b) How many different 8 bit string are there that begin and end with 10? (5)
- VI. (a) What are the properties of planar graph? (4)
 (b) Find the shortest path from node A to H using Dijkstra's algorithm. (6)



OR

- VII. (a) Define a bipartite graph and complete bipartite graph with example. (4)
 (b) Find the minimum spanning tree using Kruskal's algorithm for the given graph. (6)



- VIII. (a) Define ring, integral domain with example. (5)
 (b) Prove that $(G, *)$ is a group where $G = G_1 \times G_2$, and $(G_1, *_1)$, $(G_2, *_2)$ are groups and $*$ is defined as $(a_1, b_1) * (a_2, b_2) = (a_1 *_1 a_2, b_1 *_2 b_2)$. (5)

OR

- IX. (a) Consider the set $D100 = \{1, 2, 4, 5, 10, 20, 25, 50, 100\}$ and the partial ordering divisibility. Draw the Hasse diagram and find the supremum and infimum, upper bounds, lower bounds of the set $B = \{5, 10\}$. (5)
 (b) Prove that in a distributed lattice
 $(a \wedge b) \vee (b \wedge c) \vee (c \wedge a) = (a \vee b) \wedge (b \vee c) \wedge (c \vee a)$. (5)
